ZenteiQ AiTech Innovations

ML Engineer Assignment

Welcome. This assignment mirrors the core of what you will work on here:

Running LLM training with MaxText across different hardware and model architectures.

Work through it carefully. Your submission is the foundation of our technical discussion, so depth and clarity matter more than speed.

Reference docs: https://maxtext.readthedocs.io/en/maxtext-v0.2.2/

> *Using AI tools to understand concepts is fine.*
> *Using AI to do the work for you will be obvious in the discussion.*

Environment

Use **Google Colab**. It gives you free access to all three backends you'll need: **CPU, GPU, and TPU**. Install MaxText there across all three backend hardware and run everything in Colab.

Task 1 — Understand MaxText data formats

Before training anything, explore the data input formats MaxText supports.

Figure out:

- What formats exist and how they differ
- The tradeoffs of each

Write a short summary **in your own words**. You will reuse this knowledge in the next tasks when you set the `dataset_type` field — that value comes from what you learn here.

Extra observations from exploring the codebase are always welcome.

Task 2 — Dense model (Qwen)

Use synthetic data for this task (`dataset_type` = the synthetic option you identified in Task 1).

Start from **Qwen 0.6B** as your base model. Then create a scaled-up version — pick a larger size that your Colab hardware can actually handle. Part of the exercise is choosing a sensible target and justifying it.

Note:Train qwen 0.6B from maxtext config and submit the logs and scale it to 1B and submit log

Run	Model
A	Qwen 0.6B (base)
B	Scaled up (you choose the size)

For **both runs**, train **50 steps** on **CPU, GPU, and TPU**.

Then document, thoroughly:

NOTE: Submit all possible metrics do not be selective

- A comparison table across all three backends, for both runs
- Your interpretation:
    - Why do the numbers differ across CPU vs GPU vs TPU?
    - Why do they differ between the base and scaled-up model?
    - What exactly changed in your config to scale the model up, and why?

Starting point for your config — the rest is yours to figure out:

```
model_name: qwen3-0.6b
steps: 50
dataset_type:              # from Task 1
base_output_directory:
run_name:
```

Task 3 — MoE model (DeepSeek)

Use synthetic data for this task as well.

Take a DeepSeek MoE model and scale it **down to under 1B parameters**. Run **50 steps** on **CPU, GPU, and TPU**.

To do this you'll need to understand MoE-specific config — expert count, top-k routing, and how active vs total parameters work.

Document:

- Results across all three backends, in clear tables
- What you changed to bring the model under 1B parameters, and why
- How the MoE numbers compare to your dense (Qwen) runs, and why they differ

Final summary

Bring it all together in one consolidated writeup:

- All results, all backends, both architectures, in clear tables
- Your full interpretation: backend differences, size differences, dense vs MoE
- Anything unexpected you found along the way

> **Note:** Include *all* the metrics MaxText reports — do not pick and choose.
> Screenshots with clear captions in each part of your submission are welcome.

Submission

GitHub repository (public)

Push all your work — configs, results, metric tables, and your written analysis, including the Task 1 data format notes. Organize it however you think is best; how you structure your repo tells us how you work.

Send us your GitHub repository link.

A clear, complete submission leads to a fast and seamless next step. We'll review your repo, then reach out to schedule the discussion.